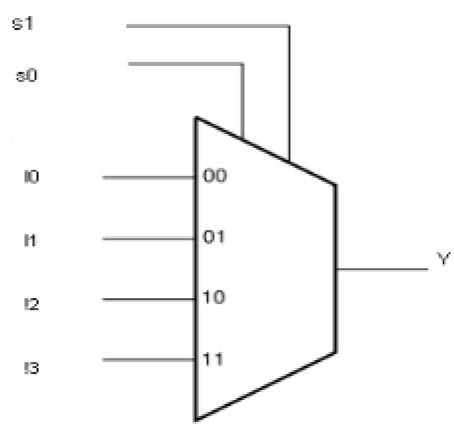
Circuit Simulation Lab:7
Multiplexers

The purpose of this experiment is to introduce the design of simple combinational circuits, in this case multiplexers. A multiplexer is a digital circuit, which is used to select a single input from the multiple input lines.



Implement the boolean expression F(A, B, C) = ∑ m(2, 3, 6, 7)

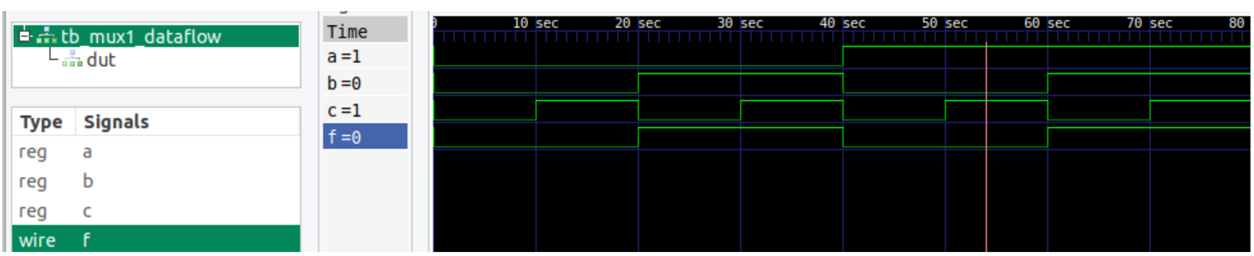
Dataflow modeling
Code:

```
module mux1_dataflow (
input a,b,c,
output f
);

assign f = (~a & b) | (a & b);

Endmodule
```

Tb:

```
module tb_mux1_dataflow;

reg a,b,c;
wire f;

mux1_dataflow dut(
.a(a),
```

```
.b(b),
.c(c),
.f(f)
);

initial begin
$dumpfile("mux1_dataflow.vcd");
$dumpvars(0,tb_mux1_dataflow);

a=0;b=0;c=0; #10;
a=0;b=0;c=1; #10;
a=0;b=1;c=0; #10;
a=0;b=1;c=1; #10;
a=1;b=0;c=0; #10;
a=1;b=0;c=1; #10;
a=1;b=1;c=0; #10;
a=1;b=1;c=1; #10;

$finish;
end

initial
$monitor("time=%0t a=%b b=%b c=%b f=%b",
        $time,a,b,c,f);

endmodule
```




```
VCD info: dumpfile mux1_dataflow.vcd opened for output.
time=0 a=0 b=0 c=0 f=0
time=10 a=0 b=0 c=1 f=0
time=20 a=0 b=1 c=0 f=1
time=30 a=0 b=1 c=1 f=1
time=40 a=1 b=0 c=0 f=0
time=50 a=1 b=0 c=1 f=0
time=60 a=1 b=1 c=0 f=1
time=70 a=1 b=1 c=1 f=1
```

Behavior modeling

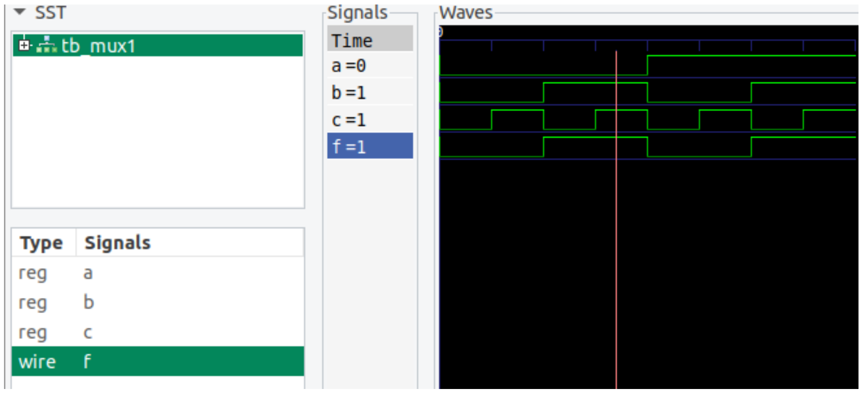
Code:

```
module mux1 (
input a , b , c ,
output reg f     );

always @(*)begin
case ({a,b})
  2'b00 : f = 0 ;
    2'b01 : f = 1 ;
      2'b10 : f = 0 ;
        2'b11 : f = 1 ;

endcase
end endmodule
```

TB:

```
module tb_mux1;
reg a , b , c ;
wire f ;


mux1 m1 (
.a(a),.b(b),.c(c),.f(f));

initial begin
$dumpfile("mux1.vcd");
$dumpvars(0,tb_mux1);
a = 0 ; b = 0 ; c = 0 ;
#10;
a = 0 ; b = 0 ; c = 1 ;
#10;
a = 0 ; b = 1 ; c = 0 ;
#10;
a = 0 ; b = 1 ; c = 1 ;
#10;
a = 1 ; b = 0 ; c = 0 ;
#10;
a = 1 ; b = 0 ; c = 1 ;
#10;
a = 1 ; b = 1 ; c = 0 ;
#10;
a = 1 ; b = 1 ; c = 1 ;
```

```
#10;
end
initial begin
$monitor ("time = %t , a = %b , b = %b , c= %b , out = %b",$time , a , b , c , f );
end
endmodule
```



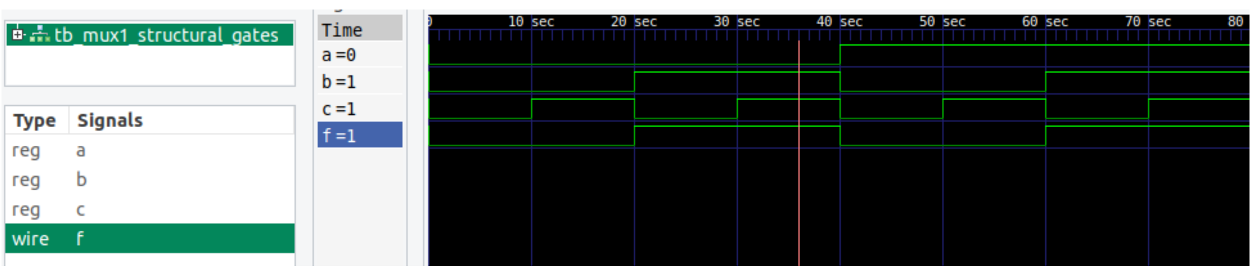

```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_7$
VCD info: dumpfile mux1.vcd opened for output.
time =                    0 , a = 0 , b = 0 , c= 0 , out = 0
time =                   10 , a = 0 , b = 0 , c= 1 , out = 0
time =                   20 , a = 0 , b = 1 , c= 0 , out = 1
time =                   30 , a = 0 , b = 1 , c= 1 , out = 1
time =                   40 , a = 1 , b = 0 , c= 0 , out = 0
time =                   50 , a = 1 , b = 0 , c= 1 , out = 0
time =                   60 , a = 1 , b = 1 , c= 0 , out = 1
time =                   70 , a = 1 , b = 1 , c= 1 , out = 1
```

Structural Modeling

Code:

```
module mux1_structural_gates(
input a,b,c,
output f
);

wire na;
wire w1,w2;

not g1(na,a);

and g2(w1,na,b);
and g3(w2,a,b);

or g4(f,w1,w2);
```

```
Endmodule

Tb:
module tb_mux1_structural_gates;

reg a,b,c;
wire f;

mux1_structural_gates dut(
.a(a),
.b(b),
.c(c),
.f(f)
);

initial begin
$dumpfile("mux1_structural_gates.vcd");
$dumpvars(0,tb_mux1_structural_gates);

a=0;b=0;c=0; #10;
a=0;b=0;c=1; #10;
a=0;b=1;c=0; #10;
a=0;b=1;c=1; #10;
a=1;b=0;c=0; #10;
a=1;b=0;c=1; #10;
a=1;b=1;c=0; #10;
a=1;b=1;c=1; #10;

$finish;
end

initial
$monitor("time=%0t a=%b b=%b c=%b f=%b",
        $time,a,b,c,f);

Endmodule
```




```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_7$ vvp out
VCD info: dumpfile mux1_structural_gates.vcd opened for output.
time=0 a=0 b=0 c=0 f=0
time=10 a=0 b=0 c=1 f=0
time=20 a=0 b=1 c=0 f=1
time=30 a=0 b=1 c=1 f=1
time=40 a=1 b=0 c=0 f=0
time=50 a=1 b=0 c=1 f=0
time=60 a=1 b=1 c=0 f=1
time=70 a=1 b=1 c=1 f=1
```

Implement the boolean expression F(A, B, C) = ∑ m(0, 1, 3, 5, 7)

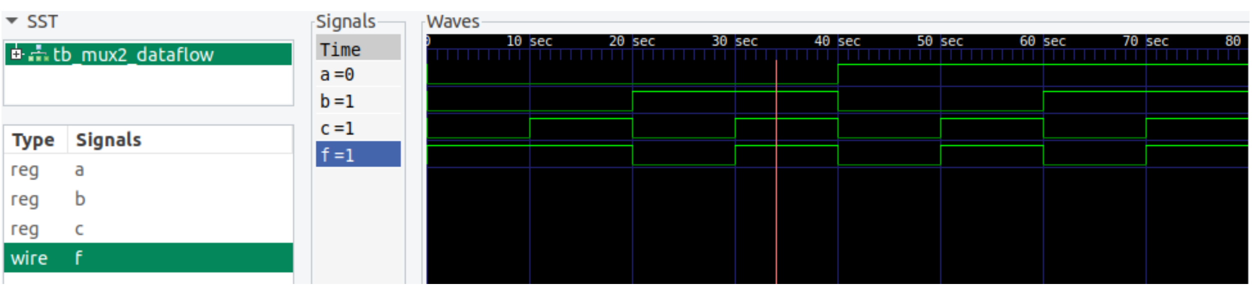
Code:

```
module mux2_dataflow (
input a,b,c,
output f
);

assign f = (~a & ~b) | c;

Endmodule
```

Tb:

```
module tb_mux2_dataflow;

reg a,b,c;
wire f;

mux2_dataflow dut(
.a(a),
.b(b),
.c(c),
.f(f)
);
```

```
initial begin
$dumpfile("mux2_dataflow.vcd");
$dumpvars(0,tb_mux2_dataflow);

a=0;b=0;c=0; #10;
a=0;b=0;c=1; #10;
a=0;b=1;c=0; #10;
a=0;b=1;c=1; #10;
a=1;b=0;c=0; #10;
a=1;b=0;c=1; #10;
a=1;b=1;c=0; #10;
a=1;b=1;c=1; #10;

$finish;
end

initial
$monitor("time=%0t a=%b b=%b c=%b f=%b",
        $time,a,b,c,f);

Endmodule
```




```
VCD info: dumpfile mux2_dataflow.vcd opened for output.
time=0 a=0 b=0 c=0 f=1
time=10 a=0 b=0 c=1 f=1
time=20 a=0 b=1 c=0 f=0
time=30 a=0 b=1 c=1 f=1
time=40 a=1 b=0 c=0 f=0
time=50 a=1 b=0 c=1 f=1
time=60 a=1 b=1 c=0 f=0
time=70 a=1 b=1 c=1 f=1
```

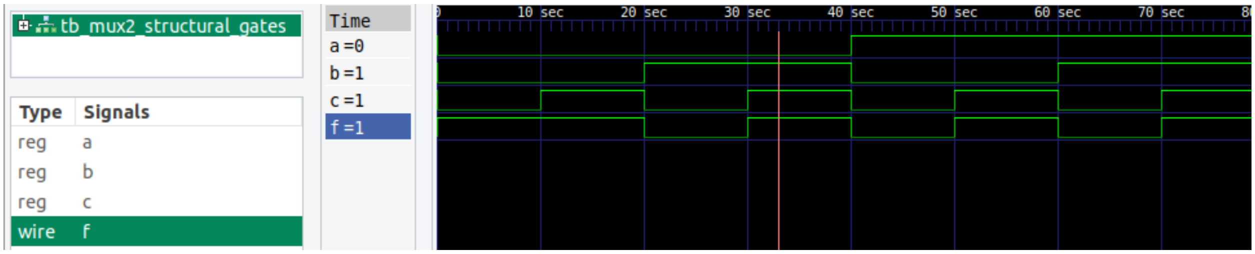
Behavior modeling

Code:

```
module mux2_structural_gates(
input a,b,c,
output f
```

```
);

wire na;
wire nb;
wire w1;

not g1(na,a);
not g2(nb,b);

and g3(w1,na,nb);

or g4(f,w1,c);

Endmodule

module tb_mux2_structural_gates;

reg a,b,c;
wire f;

mux2_structural_gates dut(
.a(a),
.b(b),
.c(c),
.f(f)
);

initial begin
$dumpfile("mux2_structural_gates.vcd");
$dumpvars(0,tb_mux2_structural_gates);

a=0;b=0;c=0; #10;
a=0;b=0;c=1; #10;
a=0;b=1;c=0; #10;
a=0;b=1;c=1; #10;
a=1;b=0;c=0; #10;
a=1;b=0;c=1; #10;
a=1;b=1;c=0; #10;
a=1;b=1;c=1; #10;

$finish;
end

initial
```

```
$monitor("time=%0t a=%b b=%b c=%b f=%b",
        $time,a,b,c,f);

endmodule
```




```
VCD info: dumpfile mux2_structural_gates.vcd opened for output.
time=0 a=0 b=0 c=0 f=1
time=10 a=0 b=0 c=1 f=1
time=20 a=0 b=1 c=0 f=0
time=30 a=0 b=1 c=1 f=1
time=40 a=1 b=0 c=0 f=0
time=50 a=1 b=0 c=1 f=1
time=60 a=1 b=1 c=0 f=0
time=70 a=1 b=1 c=1 f=1
```